Cosmic Rays and the Muon Lifetime

FYSC24

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Contents

1 Introduction **2**

2 Theory **2**

3 Detector and data taking **3**

4 Analysis **3**

5 Discussion **6**

A Setting up the experiment **8**

1 Introduction

Cosmic radiation was studied by the American physicist Carl David Anderson in 1936, and it was then that the muon was first discovered [1]. The muon is one of the elementary particles in the standard model, with a mass approximately 200 times that of the electron [2]. Despite its very short lifetime of 2.2 µs[3], the muon manages to reach Earth thanks to time dilation in the special theory of relativity. During the experiment, the muon and its decay particles, the electron and positron, will be detected, and the lifetime, i.e., the average duration of the muon's life in its rest frame, will be measured.

2 Theory

The muon belongs to the leptons in the standard model and is considered similar to the electron as it has the same charge and spin [3]. However, the mass of the muon (105 MeV) is much heavier than that of the electron, which is 0.511 MeV [2]. Cosmic particles are particles which are created from our sun, other stars and even black holes. When primary cosmic particles such as protons collide with the nuclei of atoms in the atmosphere, such as nitrogen and oxygen, new (secondary) cosmic rays are created. In turn, these secondary particles interact with new atmospheric atoms and form so-called hadronic showers. These hadronic showers spread over long distances through the atmosphere to the surface. Unstable hadrons weakly decay over time into electrons, neutrinos, and muons. Until these showers reach the ground, they may consist of electron-neutrinos, muon-neutrinos, and muons. The most common source of muons is from the decay of pions, however muons may also originate from other hadrons such as the kaon.

Muons can decay as free particles as

$$\mu^- \longrightarrow e^- + \bar{\nu}_e + \nu_\mu \tag{1}$$

$$\mu^+ \longrightarrow e^+ + \nu_e + \bar{\nu_\mu}, \tag{2}$$

which have a mean lifetime of $\tau = (2.197\,03 \pm 0.000\,04)$ µs [4]. Negative muons can also decay via muon capture as

$$\mu^- + p \longrightarrow \nu_\mu + n, \tag{3}$$

the lifetime of μ^- thus depends on in which material the muon decays, for this lab, it will be aluminium, giving the effective mean lifetime $\tau_{\mu^-} = (0.864 \pm 0.002)$ µs [4].

The exponential decay law of a particles which decays can be described as

$$N(t) = Ae^{-\lambda t}. \tag{4}$$

Above is $N(t)$ the numbers of muons at time t, A is the starting number of muons and λ is the decay factor [5]. Important to note here is that this decay only gives the muon decay. If one has multiply particles decaying one would get a series of of exponential decays happening, with their own averages lifetimes. However if these lifetimes vary from the muon lifetime can this be be seen as a constant background radiation. This would give the total number of decays as

$$N(t) = Ae^{-\lambda t} + b. \tag{5}$$

Using the decay factor λ can the mean lifetime of a muon be calculated as

$$\tau = \frac{1}{\lambda}. \tag{6}$$

3 Detector and data taking

The setup of the experiment was such that a absorber consisting of aluminium was placed between four scintillators in a dark box. Two scintillators were placed on top and two scintillators were placed below. Having two scintillators in each location made it possible to track the muons trajectories. If the two scintillators trigger simultaneously it means that the muon has indeed passed through the absorber and we can discard muons that just react externally or just passes by. The setup was such that incoming muons from cosmic rays would travel through the two upper scintillators, decay in the absorber and send out and electron which would be captured by the lower scintillators.

The scintilator works by absorbing charged particles, which excites the material of the scintilator. When the material returns from its excitation the scintilator emits a light via scintillation. At the end of each scintillator was a photomultiplier connected to convert the photons from the scintilator into an electric signal through the photoelectric effect. The photomultiplier tube worked by capturing photons on a photosensitive cathode, which in turn sends out an electron (electric signal). The electrons are multiplied through dynodes which have a electric potential. This device will thus turn single photons into a stronger electric pulse.

Each scintillator was also rapped in electrical tape, to reduce disturbance from light. Each photomultiplier was connected to a linear amplifier after that also to a discriminator. The signals from the two upper scintillators were connected to a coincidence unit, which was also done for the two lower scintillators. This coincidence unit ensured that only signals from both scintillators would result in a measurement. These were labelled up and down. More information regarding the setup can be seen in Appendix A.

To prevent the counting from particles who passes right through the absorber, the signals were coupled so the timer of the system would start when the up signal was detected along with the anti-down signal. Consequently, the timer stopped when the down signal was detected along with the anti-up signal. To prevent the two signals from not overlapping was the up, and the down, signal delayed and shortened via longer cables and a discriminator. This prevented the setup from detecting passing particles. The setup was left for 96 h to collect the relevant data.

4 Analysis

When evaluating if a result is significant or not, one can use the formula of

$$\chi^2 = \sum_{i=1}^{n} \frac{(\text{Observed values} - \text{Expected values})^2}{\text{Standard deviation}}. \tag{7}$$

By then dividing this value with the number of degrees of freedom, ν, we obtain a number which can tell us if our value is significant or not [6]. If this value is close to 1, it means that our result is statistically significant.

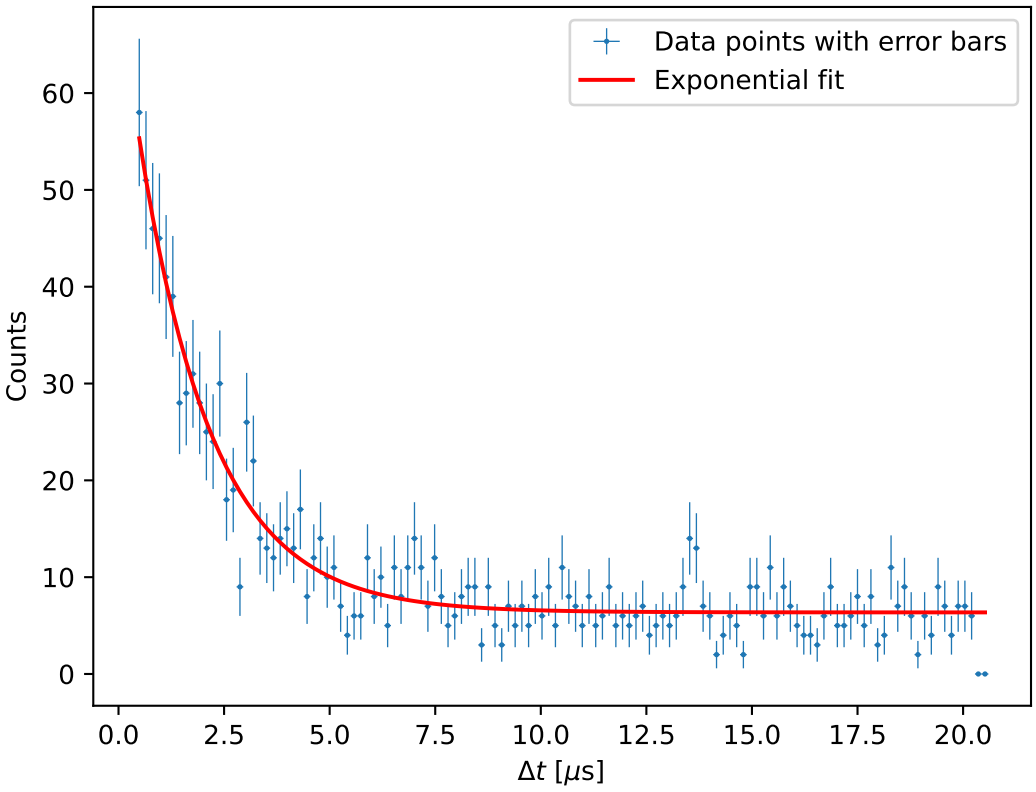


Figure 1: Plot of the muon count data and the exponential fit with $\tau = (1.74 \pm 0.10)\,\mu s$. The errors are calculated as a standard deviation of $\sigma = \sqrt{\text{Counts}}$.

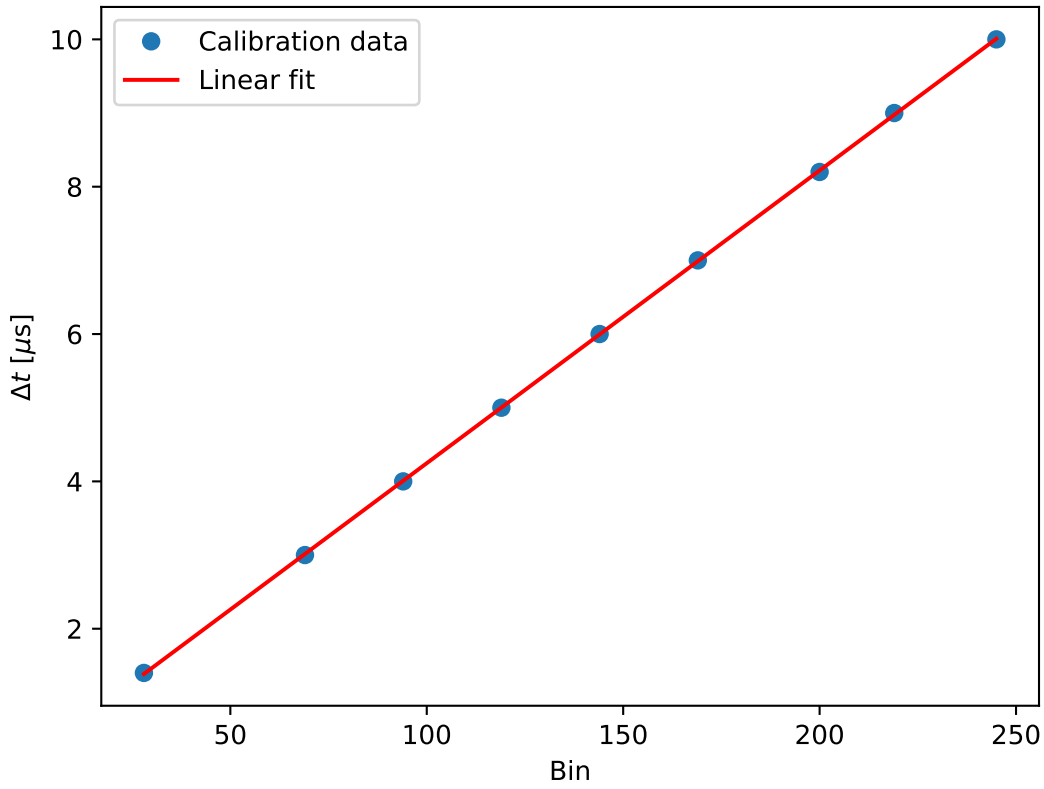


Figure 2: The linear fit to calibrate the bin numbers to corresponding Δt before the rebinning process. To obtain the data points a known time delay was used and the corresponding bin number was noted. This was repeated nine times and then a linear fit was carried out on the data points.

There is some level of data cleaning during the analysis. Firstly, the first bin were removed after rebinning and combining every four bins, since the shortest possible time the setup can measure is unknown. Due to this uncertainty, the first bin were removed to ensure that this effect didn't affect the data. Secondly, when performing the χ^2 test, bins with zero counts were removed since there otherwise would be a divide by zero error.

The function used for the fit was

$$N(t) = Ae^{-t/\tau} + b, \tag{8}$$

and the result of the fit gave a mean muon lifetime of $\tau = (1.74 \pm 0.10)\,\mu s$, and parameters $A = (65 \pm 3)$ and $b = (6.4 \pm 0.3)$. The result from the χ^2 test was $\chi^2 = 134$, $\nu = 122$ and $\chi/\nu = 1.10$, and since the reduced χ^2 is relatively close to 1 it is close to statistically significant. The calculated uncertainty in τ is calculated using the covariance matrix obtained during the fitting process and extracting the standard deviation. That is, $0.10\,\mu s$ is one standard deviation and it is only the statistical uncertainty.

One large uncertainty is the background radiation. The detector is sensitive to all charged particles, and to mitigate this effect on the measurement, discriminators are used to filter out low-energy particles. However, some background particles, like electrons, could have enough energy to trigger a start and/or stop signal in the detector.

Another uncertainty is, how the start/stop trigger works. Since it only counts as a stop if the decay product (electron/positron) exits downwards when in reality it exits omnidirectional, there is some error

in the stop trigger. The calibration of the time measurement had a very good linear fit, but it could still have some constant systematic error which was not accounted for.

5 Discussion

The measured value of the mean lifetime amounted to $\tau = (1.74 \pm 0.10)\,\mu s$. The obtained value is lower than the expected value of $\tau = 2.20\,\mu s$. The relative error for this case is 20%. This deviation is not covered by uncertainty and is much higher, as this is 4.6 standard deviations away. The deviation is significant since it can not be covered by standard deviation. The difference between the expectation value and the obtained value can be explained by the set-up not differentiating between positive and negative muons. Since negative muons have a much shorter effective lifetime in aluminium of $\tau_{\mu^-} = (0.864 \pm 0.002)\,\mu s$, due to muon capture, it will lower the average. That is, the experiment does not account for different decay dynamics between muons in a vacuum and muons interacting with matter [7]. Taking a simple average of the different types of decay, the mean lifetime should be $\tau_{\text{mean}} = 1.53\,\mu s$, which is closer but not correct. However, this average assumes an equal amount of both types of muons, but since negative muons have a shorter decay time in materials due to muon capture it is not unreasonable that there are fewer negative muons entering the detector. If the 35 % of the muons in the detector are μ^- the weighted mean would be $\tau_{\text{mean}} = 1.73\,\mu s$ which is close to the experimental value. It should be noted that these calculations are crude and are only used to discuss a possible reason why the value differs. Additionally, other particles, mainly electrons, are introduced to the system which can hinder the calculations further and are the biggest source of the systematic error.

While reducing the total uncertainty is a good method, in cases where the results deviate from the expected value so significantly, the more feasible solution could be reconsidering the standard deviation to better capture uncertainties. Another method could be setting up the experiment again to see if the same error persists. One more approach is measuring the typical background noise that gets captured in the process and subtracting it from the final results and thanks to that both results and uncertainty could be improved.

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A Setting up the experiment

1. In this experiment must one account for all the particles that interfere with this setup. The particles that do interfere are all particles that react via the electromagnetic force. These include all charged particles. These may come from cosmic rays or nearby radiation sources. These can be handled by accounting for the incoming particles which do not interfere with the absorber. The particles which interfere with our absorber, which we do not want, should be almost constant. Hence can these be removed afterwards.

2. The width in the picture below is the timescale of the signal in the scintillator from the particle. Since the mean lifetime of the muon is around 2.2 µs, we should use a timescale smaller than this [3]. In Figure 3 we can see that this timescale is on the order of nano seconds, which is good.

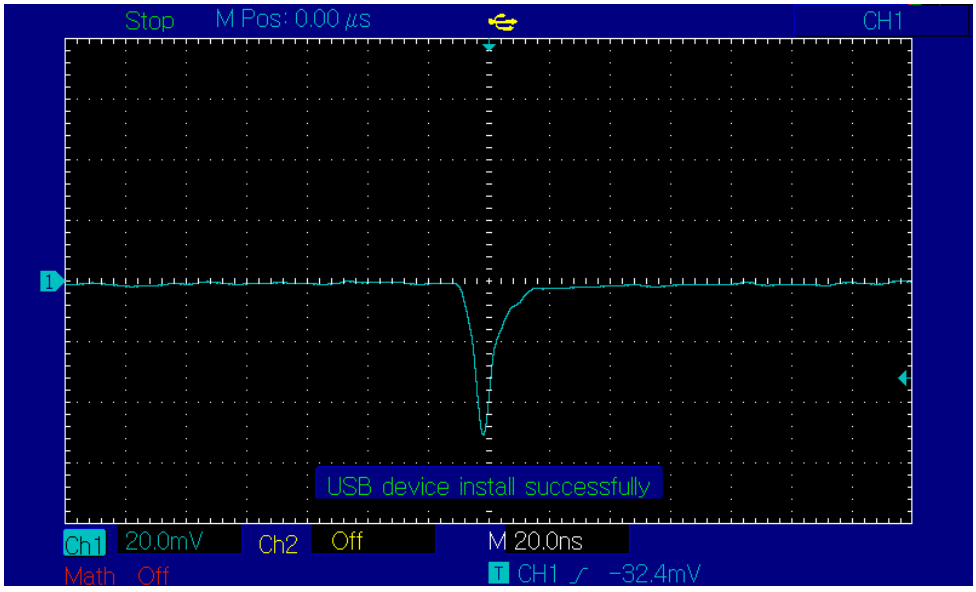


Figure 3: This picture shows the linear amplitude displayed on an oscilloscope for the first scintillator.

3. After the signal has passed through the discriminator it has the shaped seen in Figure 4. Here can one see that the timescale is on the order of 20 ns.

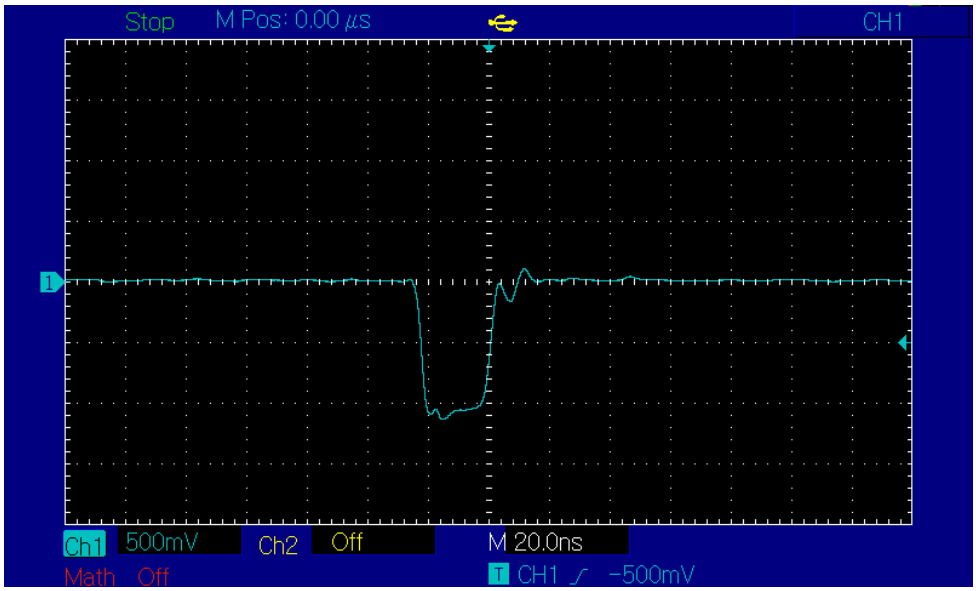


Figure 4: This picture shows the signal after the discriminator on an oscilloscope for the first scintillator.

4. When viewing the discriminated signal together with the non discriminated signal could the threshold on the non discriminated signal be changed. When the trigger level was lower then 20 mV disappeared the discriminated signal sometimes. This is because the discriminator only accepts signal of a certain strength. By checking the threshold for all signal was it found that the thresholds were somewhere between 20-30 mV. When the trigger level on the discriminated signal was lowered, were no signal visible.

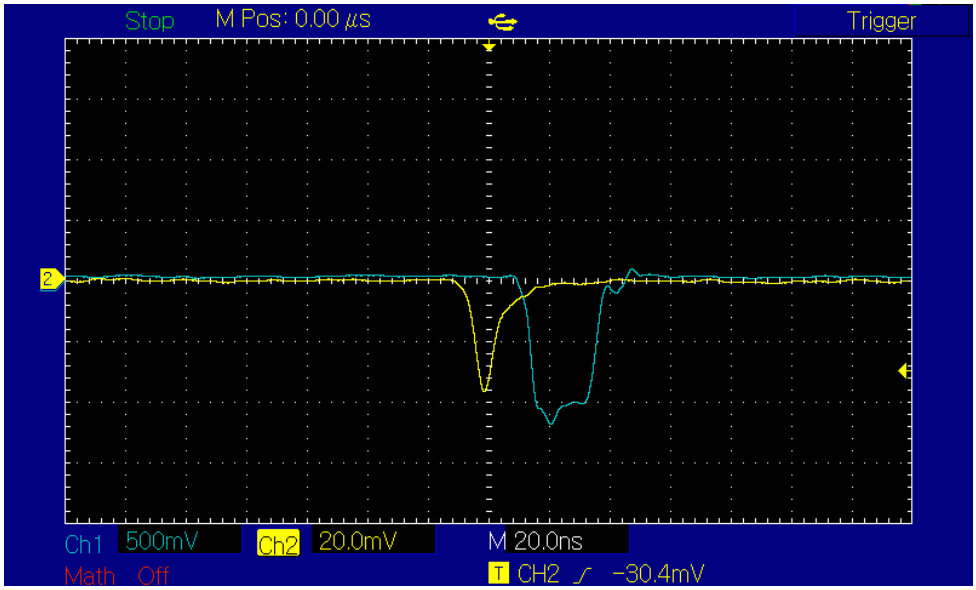


Figure 5: This picture shows both signals on an oscilloscope for the first scintillator.

5. Some particles just passes through the setup and is not wanted in the experiment. To eliminate these was the signals coupled so that the timer would only start if the upper scintillator were activated at the same time as the ant-down signal was activated. This would yield the output in Figure 6.

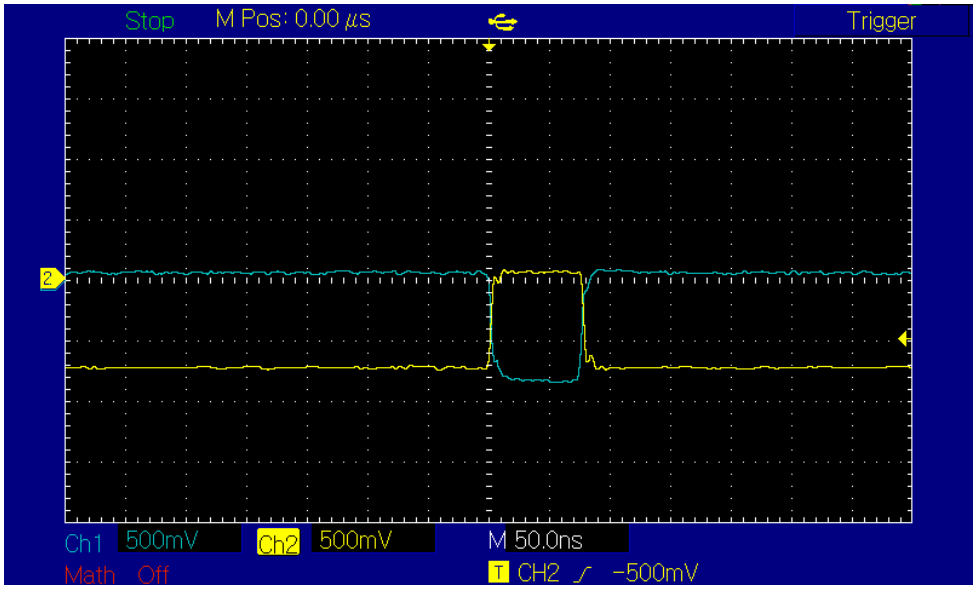


Figure 6: This picture shows how it should look when the timer starts.

6. However there would sometimes be a short delay between the signals in the case when a particle passes right through. Also sometimes would it also be a delay between the up and anti-down signal. This had to be accounted for. These had to be accounted for by delaying and shorting the up- and down-signal whilst shortening them. This yielded signals which can bee seen in Figure 7.

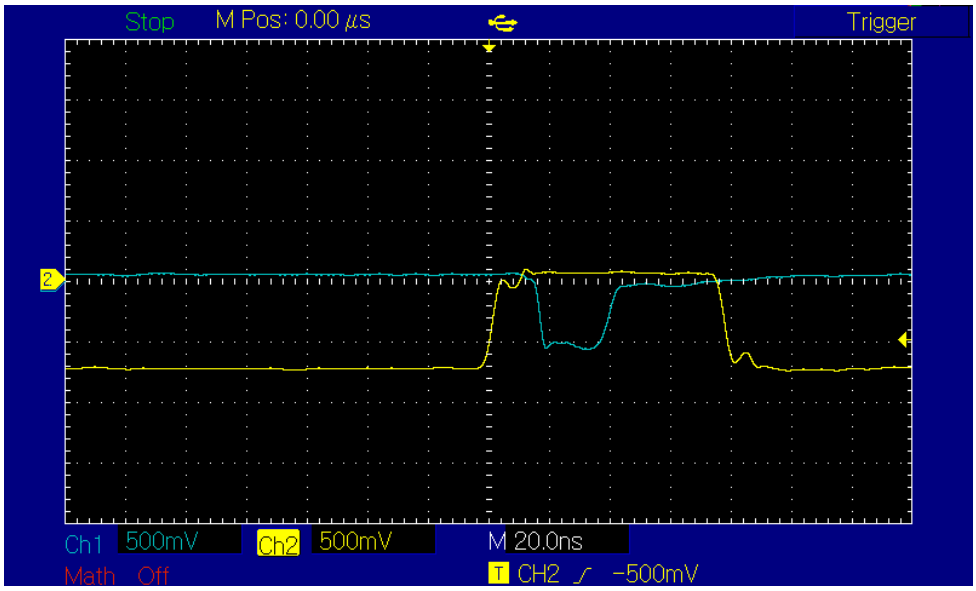


Figure 7: This picture shows the up signal and anti-down signal when the up has been shortened and delayed. This made it easier for the system to not count for particles just passing through. The same was done for the down and anti-up signal.